SR NEXT INTERNSHIP PROGRAM

AI-Powered Instagram Influencer Analytics & Automation Insights

Week 1 Report – Instagram Influencer Data Collection and Dataset Preparation

Project Title: AI-Powered Instagram Influencer Analytics & Automation Insights
Internship Program: SR NEXT Data Science & AI Internship
Report: Week 1 Report (Phase 1 Deliverable)
Candidate / Intern: Data Science & AI Intern
Project Phase: Phase 1 – Week 1 (Data Collection & Quality Control)
Submission Date: October 2026
Target Cohort: Micro & Mid-Tier Public Instagram Influencers (< 1,000,000 Followers)

✔ DELIVERABLE STATUS: SUBMISSION-READY (2,561 VALIDATED PROFILES)

2. Executive Summary

This report documents the completion of Week 1 for the **AI-Powered Instagram Influencer Analytics & Automation Insights** internship project at **SR NEXT**. The fundamental mission of Phase 1 was to collect authentic public Instagram creator data and construct a clean, structured, and submission-ready dataset comprising more than 2,000 profiles with follower counts strictly below 1,000,000.

Following an initial technical investigation of direct Instagram access, automated live crawling was determined to be technically throttled and blocked by Instagram's bot defenses (HTTP 401/429 rate limits, login challenges, and bot detection perimeters). In accordance with the internship's ethical standards—which prohibit credential stuffing, session hijacking, or CAPTCHA circumvention—data collection transitioned to legitimate, publicly downloadable open-source Instagram dataset repositories hosted on GitHub.

By integrating two authentic open-data archives from Bright Data (luminati-io), the data engineering pipeline assembled a merged baseline of 3,129 authentic creator profiles. A comprehensive quality-control pass then executed dual-stage linguistic filtering (pruning 568 foreign-language dominant accounts to focus exclusively on English-dominant creators), stripped all Unicode emojis from designated textual fields, normalized hashtags into uniform `#tag1;`

`#tag2` structures, standardized content themes into comma-delimited lists, mapped niches against the Phase 1 taxonomy, and validated engagement rate calculations. The final deliverable comprises **2,561 clean, authentic profiles**, achieving full schema parity across CSV and XLSX formats with zero missing core fields and 100% data provenance.

3. Week 1 Objective

The objective of Week 1 was to deliver an empirical foundation of verified public Instagram influencer data for downstream analytical tasks, such as creator clustering, reach modeling, and automation classification in Phase 2.

The core internship specifications mandated:

- **Volume Target:** Prepare a structured dataset containing *more than 2,000* unique public profiles.
- **Audience Constraint:** Limit profiles strictly to *followers < 1,000,000* (nano, micro, and mid-tier/macro tiers).
- **Authenticity Requirement:** Utilize 100% authentic public data with zero synthetic, fabricated, or mocked records.
- **Standardized Schema:** Capture 17 essential dimensions covering creator identity, engagement metrics, posting cadence, content taxonomy, and automation markers.

Required Internship Data Fields

- **Name / Username:** Public display name and canonical Instagram username handle.
- **Followers:** Public audience count strictly bounded between 1 and 999,999.
- **Instagram Handle:** Normalized identifier with leading '@' symbol (e.g., `@username`).
- **Category / Niche:** Standardized classification matching the approved 18-niche taxonomy + Other.
- **Engagement Rate:** Mathematically verified interaction ratio: `(Likes + Comments) / Followers * 100`.
- **Public Contact Details:** Business email address or phone number when publicly shared, or standard fallback.
- **Posting Frequency:** Estimated monthly post volume calculated from post dates and publishing activity.
- **Hashtags & Content Themes:** Clean semicolon-delimited hashtags and comma-delimited conceptual topics.
- **Instagram Automation Usage:** Heuristic indicator evaluating activity

regularity (Possible, Unlikely, Unknown).

- **Additional Notes:** Concise qualitative observations regarding verification and engagement outliers.

4. Data Collection Process

The collection process followed a rigorous, multi-stage engineering workflow:

1. **Requirements Analysis & Compliance Framing:** Reviewed project requirements, field schemas, and ethical data collection boundaries.
2. **Investigation of Public Instagram Access:** Tested automated unauthenticated HTTP requests (Requests, BeautifulSoup, Instaloader). Observed that Instagram blocks unauthenticated scrapers via HTTP 401/429 status codes and mandatory login barriers. Direct scraping was halted immediately without attempting credential stuffing or CAPTCHA circumvention.
3. **Transition to Public Open-Source Data:** Identified legitimate, open-source Instagram dataset repositories hosted on GitHub by Bright Data (luminati-io) for academic and benchmarking purposes.
4. **Data Ingestion & Quarantine:** Downloaded raw open-source files into `data/raw/`: an initial 1,000-post multi-niche sample and a 2,217-creator coding influencer spreadsheet.
5. **Schema Mapping & Normalization:** Engineered ingestion modules (`src/collect.py`, `src/import_secondary_source.py`) to harmonize disparate schemas into the unified 17-column specification.
6. **Dataset Merging & Deduplication:** Combined 949 valid base records and 2,180 secondary records, producing a baseline of 3,129 unique accounts with zero duplicate handles.
7. **Boundary Verification:** Programmatically verified that all candidate records complied with the `followers < 1,000,000` constraint.
8. **Quality Control & Text Normalization:** Executed `src/final_clean_qc.py` to filter foreign-language dominant profiles, strip emojis, normalize hashtags and themes, standardize categories, and clean contact information.
9. **Validation & Deliverable Export:** Executed automated quality validation and exported twin deliverables in CSV and native OpenXML XLSX format.

5. Tools and Technologies Used

The project relied entirely on standard data engineering tools and Python libraries, ensuring transparent reproducibility:

Python 3.13
Programming Language
Core execution runtime for data ingestion, cleaning, transformation, and validation scripting.
Pandas 3.0.2
Data Processing Library
Tabular data engine utilized for dataframe manipulation, schema mapping, deduplication, and aggregation.
NumPy 2.3.3
Numerical Computing
Vectorized computation of mathematical engagement rates and numerical boundary auditing.
OpenXML (ZIP/XML)
Document Generation
Engineered native generation of XLSX spreadsheets and DOCX reports without external binary C-dependencies.
CSV (RFC 4180)
Tabular Data Format
Standard deliverable format for machine-readable storage, ensuring universal cross-platform compatibility.
Antigravity
AI Assistant Environment
Interactive engineering and development environment used for auditing, verification, and pipeline automation.
Git & GitHub
Version Control & Data Hub
Used to track project source files and retrieve public open-source benchmark dataset repositories.
Note: Browser automation tools such as Selenium, Puppeteer, Playwright, or unauthenticated scraping bots were explicitly NOT used for final data collection.

6. Data Sources

To satisfy the quantitative threshold of > 2,000 profiles while upholding platform terms, the final dataset combined two authentic public datasets:

Repository Name
luminati-io/Instagram-Posts-dataset-samples
luminati-io/Instagram-dataset-samples
Repository URL
https://github.com/luminati-io/Instagram-Posts-dataset-samples
https://github.com/luminati-io/Instagram-dataset-samples
Source File Name
Instagram - Posts.csv
Instagram Profiles - Github Hashtag.xlsx
Direct Raw URL
https://raw.githubusercontent.com/.../Instagram%20-%20Posts.csv
https://raw.githubusercontent.com/.../Instagram%20Profiles%20-

%20Github%20Hashtag.xlsx
Local Raw File
data/raw/instagram_profiles_raw.csv
data/raw/instagram_coding_influencers_raw.xlsx
Niche Focus
Multi-category (Lifestyle, Art, Fashion, Travel, etc.)
Technology, Coding, Software Engineering, Developers
Initial Raw Records
1,000 raw post records
2,217 raw creator records
Valid Imported Profiles
949 unique profiles (< 1M followers)
2,180 unique profiles (< 1M followers)
Final Clean Profiles
613 profiles (23.94% of final deliverable)
1,948 profiles (76.06% of final deliverable)
Provenance Column Tag
BrightData Public Instagram Dataset (GitHub)
BrightData Public Instagram Coding Influencers Dataset (GitHub)

Provenance & Ethical Justification

1. Strict Provenance Tracking: Each row explicitly records its origin in the `data_source` column. No profiles were blended or fabricated.
2. Platform Compliance: Automated unauthenticated scraping violates Instagram's Terms of Service. Sourcing verified public datasets maintained legal and ethical compliance.
3. Complete Reproducibility: Both source repositories are publicly accessible on GitHub, allowing full end-to-end auditability.

7. Dataset Schema

The final dataset strictly implements the 17-column specification required for Phase 1:

1
`profile_url`
String
Canonical profile URL (https://www.instagram.com/<username>/).
2
`name`
String
Display name with emojis stripped; falls back to username if blank.
3
`username`
String
Unique Instagram handle identifier.
4
`instagram_handle`
String

Formatted handle with leading '@' symbol (e.g., @creator).
5
`followers`
Integer
Total follower count, strictly bounded: 1 ≤ followers ≤ 982,005.
6
`category`
String
Creator niche categorized according to the Phase 1 18-niche taxonomy + Other.
7
`avg_likes_last_10`
Float
Average likes per post as reported by source dataset.
8
`avg_comments_last_10`
Float
Average comments per post as reported by source dataset.
9
`engagement_rate`
Float
Computed percentage: (avg_likes + avg_comments) / followers * 100.
10
`posting_frequency_per_month`
Float
Estimated monthly posts derived from post timestamps and activity cadence.
11
`hashtags`
String
Cleaned, semicolon-delimited hashtag string (#tag1; #tag2).
12
`content_themes`
String
Cleaned, comma-delimited conceptual themes (Theme1, Theme2).
13
`contact_details`
String
Public email or phone number, or standardized fallback 'Not publicly available'.
14
`automation_usage`
String
Heuristic assessment flag: 'Possible', 'Unlikely', or 'Unknown'.
15
`additional_notes`
String
Concise observations (e.g., 'Verified account', 'Low engagement rate').
16
`data_source`
String
Explicit provenance tracking tag identifying the originating GitHub repository.
17
`collection_date`
String
ISO 8601 UTC timestamp recording ingestion and processing time.

8. Data Cleaning and Preprocessing

A comprehensive cleaning pass (`src/final_clean_qc.py`) prepared the submission-ready dataset:

- **Deduplication:** Confirmed zero duplicate usernames, handles, or profile URLs across all merged records.
- **Follower Bounds:** Programmatically verified that 100% of candidate profiles satisfied `0 < followers < 1,000,000`.
- **English-Dominant Linguistic Filter:** Evaluated profile bios, names, hashtags, and themes. Removed 325 accounts dominated by non-Latin scripts (Cyrillic, Arabic, CJK, Devanagari, Thai) and 243 accounts dominated by foreign Latin stopwords (Portuguese, Spanish, Turkish, Indonesian, German, French, Italian). A total of 568 non-English profiles were pruned, leaving 2,561 clean English-dominant creators.
- **Unicode Emoji Stripping:** Applied NFKC normalization and multi-block Unicode regex to eliminate emojis from all textual fields (`name`, `category`, `hashtags`, `content_themes`, `contact_details`, `additional_notes`).
- **Hashtag Normalization:** Parsed JSON arrays, stripped brackets/quotes, ensured uniform '#' prefixes, removed duplicates within records, and joined tokens with semicolons: `#tag1; #tag2`.
- **Content Theme Normalization:** Stripped brackets, quotes, and '#' prefixes, capitalized initial letters, removed duplicates, and formatted themes as comma-separated lists: `Theme1, Theme2`.
- **Category Taxonomy Alignment:** Verified all category strings against the approved 18-niche taxonomy + Other. Confirmed zero unmapped category labels.
- **Contact Details Standardization:** Preserved valid emails and phones while standardizing missing or empty values to `Not publicly available`.

9. Dataset Validation

Quality-control scripts (`src/validate.py` and `src/final_clean_qc.py`) verified compliance across all dimensions:

Total Profile Count
> 2,000 profiles
2,561 profiles
PASSED (+561)
Target Window
Approx. 2,400–2,600
2,561 profiles
MET

Unique Usernames
100% unique
2,561 unique handles
PASSED
Duplicate Count
0 allowed
0 duplicates
PASSED
Minimum Followers
Followers > 0
1 follower
PASSED
Maximum Followers
Followers < 1,000,000
982,005 followers
PASSED
Followers ≥ 1M
0 allowed
0 accounts
PASSED
Followers ≤ 0
0 allowed
0 accounts
PASSED
Missing Core Fields
0% nulls in identity/reach
0 nulls
PASSED
Total Schema Columns
Exactly 17 columns
17 columns
PASSED
CSV / XLSX Parity
Identical rows & values
100% cell-by-cell match
PASSED
Emoji Count in Text
0 emojis in target fields
0 emojis detected
PASSED

Summary Statistics of Numerical Attributes

Followers
1
272
708
16,985.49
3,532
982,005

Engagement Rate (%)
0.00%
1.59%
5.21%
11.53%
11.90%
4,400.00%
Posting Frequency (/mo)
0.00
4.15
8.33
13.32
24.50
30.00

Niche Distribution Breakdown

Technology
1,542
60.21%
Other
473
18.47%
Business & Finance
113
4.41%
Fashion & Beauty
85
3.32%
Art & Design
71
2.77%
Education
47
1.84%
Travel
36
1.41%
Lifestyle
35
1.37%
Fitness & Health
29
1.13%
Food & Cooking / Photography
56 (28 each)
2.18%
Remaining 7 Niches
74
2.89%

10. Engagement Rate

Creator engagement rate was computed according to the formula specified by

the internship:

```
Engagement Rate (%) = ((Average Likes + Average Comments) / Followers) * 100
```

Numerical validation confirmed 100% mathematical consistency with this formula across all 2,561 records (max diff ≤ 1e-4).

Data-Source Limitation Disclosure

The technical audit identified a critical nuance regarding the fields `avg_likes_last_10` and `avg_comments_last_10`:

- These figures represent creator interaction averages provided by the originating open-source research dataset.
- Because Instagram's live feed was not directly crawled due to anti-scraping blocks, the pipeline did not crawl 10 individual post endpoints per creator.
- Rather than simulating or fabricating post-level numbers, the pipeline transparently mapped the authentic source metrics into the schema, preserving data provenance and integrity.

11. Hashtags and Content Themes

Hashtags and content themes provide essential semantic features for creator segmentation. The pipeline applied structured standardization rules:

- Removed brackets, quotes, and all Unicode emojis.
- Standardized hashtags into clean semicolon-delimited strings: `#tag1; #tag2; #tag3`.
- Extracted conceptual topics, title-cased tokens, and formatted themes as comma-delimited strings: `Theme1, Theme2, Theme3`.
- Deduplicated tokens case-insensitively while preserving original creator terminology without lossy machine translation.

12. Limitations

In the interest of technical accuracy and academic rigor, several limitations are acknowledged:

- **Platform Anti-Scraping Defenses:** Instagram actively defends its perimeter against unauthenticated bots, preventing live automated crawling without violating platform policies.
- **Open-Source Sample Reliance:** The dataset is derived from public research archives. While authentic, updates depend on open-data release cycles.
- **Post-Level Granularity:** Individual post-level timestamps and comment

text bodies were not available across all profiles, limiting granular time-series modeling.

- **Deferred Optional Metrics:** Market Share and Influence Score were not present in the raw data and have been transparently deferred to Phase 2 algorithmic modeling rather than fabricated.
- **Contact Details Sparsity:** Public contact info was present for 276 profiles (10.78%), with the remainder properly marked as `Not publicly available`.

13. Final Deliverables

The complete suite of Phase 1 Week 1 deliverables has been generated and validated:

`instagram_influencers_final_clean.csv`
CSV (RFC 4180)
2,561 rows (1.2 MB)
Primary clean tabular deliverable, 17 columns.
`instagram_influencers_final_clean.xlsx`
Excel (OpenXML)
2,561 rows (354 KB)
Clean spreadsheet deliverable with full schema parity.
`Week_1_Report.pdf`
PDF Document
Multi-Page Document
Formal executive and technical internship report.
`Week_1_Report.docx`
Word (DOCX)
Editable OpenXML
Submission-ready editable document of the report.
`Week_1_Report_Data_Sources.txt`
Plain Text
Complete Index
Detailed provenance record of GitHub repositories and URLs.
`final_cleaning_report.txt`
Text QA Log
5.5 KB
14-section empirical quality-control audit log.

14. Conclusion

Week 1 of the SR NEXT Internship successfully fulfilled all Phase 1 mandates. Through ethical data sourcing, automated data engineering, and multi-stage quality control, the project produced a clean, standardized, and validated dataset of **2,561 authentic Instagram creator profiles**. With strict follower boundaries (< 1M), zero duplicates, verified engagement rate calculations, and an English-dominant focus, the dataset provides an ideal foundation for Phase

2 exploratory data analysis, creator clustering, and automation insights.